

INTELLIGENT LIBRARY ASSISTANCE BOT

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Abstract- Modern library management faces several challenges in locating and retrieving books efficiently, particularly in large institutions where book organization and user service are time-consuming and labour-intensive. To address these issues, an Intelligent Library Assistance Bot is proposed — a fully automated robotic system capable of locating, identifying, and retrieving requested books with minimal human intervention. The system integrates computer vision, QR-code-based navigation, IoT connectivity, and robotic arm manipulation to automate the process of book retrieval and delivery.

The proposed system is powered by a Raspberry Pi 4 acting as the main controller and an ESP32 microcontroller for low-level motor and sensor management. Navigation within the library is achieved through QR-coded path tracking and ultrasonic sensors for obstacle avoidance. The robot identifies shelves through camera-based QR scanning and retrieves books using a servo-controlled gripper mechanism. Furthermore, it incorporates cloud synchronization using Google AppSheet and Google Sheets, ensuring real-time tracking of requests and operational data. This system minimizes human workload, reduces search time, and enhances operational efficiency. The integration of automation and intelligent control within a cost-effective hardware design demonstrates a scalable approach to transforming traditional libraries into smart, autonomous, and technology-driven environments.

Keyword- Library Automation, QR Code Navigation, Raspberry Pi, Computer Vision, Cloud Integration, Robotic Arm.

I. INTRODUCTION

Libraries have long served as central repositories of knowledge and information, enabling people to access, borrow, and study various academic and literary resources. However, as institutional libraries continue to expand, managing and retrieving books efficiently has become increasingly complex. Manual searching and retrieval not only consume considerable time but also lead to operational inefficiencies and errors, particularly when dealing with thousands of volumes. In conventional systems, users depend on librarians to locate books manually, which results in significant delays and limited scalability. In the age of digital transformation, automation technologies and robotics are redefining the way libraries operate. While traditional automation methods like **RFID-based tagging** and **barcode management** have simplified cataloguing and tracking, they do not address the **physical retrieval** of books — which remains a labour-intensive task. Recent advances in **robotics**, **artificial intelligence (AI)**, and the **Internet of Things (IoT)** have opened new opportunities for developing autonomous systems capable of performing precise, repetitive, and complex physical operations. The **Intelligent Library Assistance Bot** bridges this gap by combining automation with real-time navigation and computer vision to perform autonomous book retrieval. The bot navigates using a QR-code-based guidance system printed on the floor, scans shelves to identify target books, and uses a robotic arm equipped with a gripper to retrieve and deliver the books to a designated location. The use of a **Raspberry Pi 4 microprocessor** for high-level operations and an **ESP32 microcontroller** for real-time actuation and sensing ensures reliable performance and modular scalability. This integration enables the bot to interpret visual

markers, process commands, and communicate with a cloud-based database to manage requests and maintain transaction logs. Libraries that deploy this system can expect **reduced human effort, faster response times, and improved user satisfaction**. The project also aligns with the broader trend of **smart infrastructure development**, showcasing how embedded systems and robotics can modernize traditional educational environments.

The proposed system's design philosophy focuses on **affordability, modularity, and accuracy**. It uses widely available hardware components such as IR sensors, DC motors, and servo actuators, reducing costs while maintaining high precision in navigation and control. Compared to industrial Automated Storage and Retrieval Systems (ASRS), the Intelligent Library Assistance Bot provides a **low-cost yet effective** solution that educational institutions can readily implement. The following sections of this paper discuss the **existing system limitations, objectives of the proposed design, and the technical methodology** used in implementing the Intelligent Library Assistance Bot. Additionally, the paper elaborates on the hardware and software integration, algorithm design, and performance evaluation, followed by discussions on results, conclusions, and potential future developments.

II. EXISTING SYSTEM

In conventional library systems, the process of locating, retrieving, and issuing books primarily relies on manual human effort. Librarians or library assistants are responsible for identifying the requested book's location using a digital or physical catalogue and then searching through numerous racks and shelves to retrieve it. This procedure, though traditional, is time-consuming and prone to errors, especially in large-scale institutional libraries where the number of books and their frequent rearrangement make manual searching inefficient. Existing digital solutions have introduced certain levels of automation, such as barcode scanning and Radio Frequency Identification (RFID) technology. These systems enable quicker book identification and streamline the issuing process. However, these technologies only manage data retrieval and book tracking, not physical automation. That is, they can confirm whether a book exists in the library and its location in the database, but they cannot autonomously fetch or deliver it.

Moreover, existing systems often fail to provide real-time user updates, cloud synchronization, or physical mobility within the library environment. In many cases, even with digital catalogues, users must manually locate the shelf, climb ladders, and sort through tightly packed

books — an effort-intensive process that can lead to delays, especially during high-demand hours.

In addition, traditional systems lack interactivity and adaptability. If a book is misplaced, manually catalogued incorrectly, or physically obstructed, the system cannot detect or correct the error automatically. The absence of integrated sensors and autonomous mobility further limits scalability, making such systems inefficient for smart campus environments. Thus, there exists a clear need for a robotic solution that combines physical automation, machine intelligence, and cloud-based data management. The Intelligent Library Assistance Bot addresses this limitation by merging QR navigation, computer vision, and IoT connectivity to automate the complete workflow of book retrieval — from request submission to delivery.

III. OBJECTIVE

The primary goal of the Intelligent Library Assistance Bot is to design and implement a fully automated system that can locate, identify, and retrieve books autonomously with high precision and minimal human intervention. The objectives of the proposed system are as follows:

1. To automate book retrieval:

Replace manual searching and physical book collection processes with an autonomous robotic system capable of navigating through library aisles and retrieving books.

2. To enhance operational efficiency:

Reduce the time required for locating and issuing books by using QR code-based path tracking and real-time communication between the robot and a centralized database.

3. To minimize human effort and errors:

Limit human involvement in repetitive tasks such as searching and sorting books, thereby reducing the risk of misplacement and improving overall accuracy.

4. To integrate IoT-based cloud connectivity:

Enable data synchronization using platforms such as Google Sheets and AppSheet, allowing real-time updates, remote monitoring, and efficient request management.

5. To ensure cost-effectiveness and scalability:

Develop the system using low-cost and readily available hardware components (Raspberry Pi, ESP32, IR sensors, and DC motors), ensuring easy scalability for libraries of varying sizes.

6. To improve user experience:

Deliver faster response times and automated service, allowing users to request books digitally

and receive them without manually searching through racks.

7. To promote the adoption of robotics in education:

Demonstrate how robotic and embedded system integration can contribute to smart educational infrastructures and encourage technological learning environments.

Through these objectives, the Intelligent Library Assistance Bot aims to bridge the gap between digital cataloguing systems and physical automation, creating a complete end-to-end smart library experience.

IV. PROPOSED DESIGN

The Intelligent Library Assistance Bot is designed as a modular robotic system that integrates navigation, identification, retrieval, and delivery mechanisms into a cohesive workflow. The design ensures high accuracy in locating books, reliable performance in movement, and safe handling during retrieval.

A. System Overview

The robot comprises four primary functional modules:

Navigation Module – responsible for autonomous movement across library aisles using QR-coded floor markers and ultrasonic sensors for obstacle detection.

Book Identification Module – uses a camera and QR scanner mounted on the robotic arm to identify shelf labels and specific book codes.

Retrieval Mechanism – consists of a servo-controlled robotic arm and gripper to pick up the required book.

Delivery System – manages the safe transportation of the retrieved book to the librarian's desk or user collection point.

These modules are interconnected through a Raspberry Pi–ESP32 communication architecture. The Raspberry Pi 4 performs high-level image processing, navigation logic, and cloud connectivity, while the ESP32 controls the actuators and sensors at the hardware level.

B. Hardware Components

Raspberry Pi 4 – Acts as the main controller responsible for processing image data, scanning QR codes, managing cloud connectivity, and coordinating high-level commands.

ESP32 Microcontroller – Handles sensor inputs and motor actuation; serves as an intermediary between the Raspberry Pi and peripheral devices.

Camera Module – Used for capturing QR codes placed on shelves and floor markers for navigation.

IR and Ultrasonic Sensors – Provide path detection, distance measurement, and obstacle avoidance functionality.

Servo and DC Motors – Enable movement and control of the robotic arm and gripper for precise book handling.

Motor Driver (L298N) – Controls direction and speed of the motors based on navigation instructions from the microcontroller.

LCD Display – Displays system status, messages, and navigation data for user feedback.

Power Supply – Rechargeable Li-ion batteries power the bot, ensuring mobility without external wiring.

C. Software Components

The software subsystem integrates several layers of control:

Operating System: Raspberry Pi OS running Python-based control scripts.

Programming Languages: Python and C++ for control logic, OpenCV for computer vision, and Arduino IDE for ESP32 programming.

Database and Cloud Integration: Google Sheets for request logging and AppSheet for front-end book request management.

Communication Protocols: Serial communication (UART) between Raspberry Pi and ESP32; Wi-Fi for cloud connectivity.

D. System Operation Flow

When a user requests a book through the cloud interface, the system searches its digital catalogue to identify the corresponding rack and row ID. The Raspberry Pi receives this data and initiates navigation toward the correct rack by scanning QR-coded floor markers. Once the target location is reached, the robotic arm's camera scans the rack's QR code and matches it to the requested book. Upon identification, the arm extends toward the shelf, grips the book securely, and places it in a book tray mounted on the bot's base. The robot then returns along the same path to deliver the book at the librarian's counter. This process ensures complete automation from request to delivery, eliminating manual searching and retrieval.

E. Design Features and Benefits

High Precision Navigation: QR-coded markers provide low-cost yet reliable path tracking without complex sensors like LIDAR.

Safe Book Handling: Servo-controlled gripper ensures delicate operation to prevent book damage.

Scalable Architecture: Modular hardware and cloud integration enable easy system expansion.

Real-Time Monitoring: Cloud synchronization ensures librarians can monitor ongoing tasks, queue status, and completed deliveries.

User-Friendly Interface: Requests can be submitted through any internet-connected device via AppSheet.

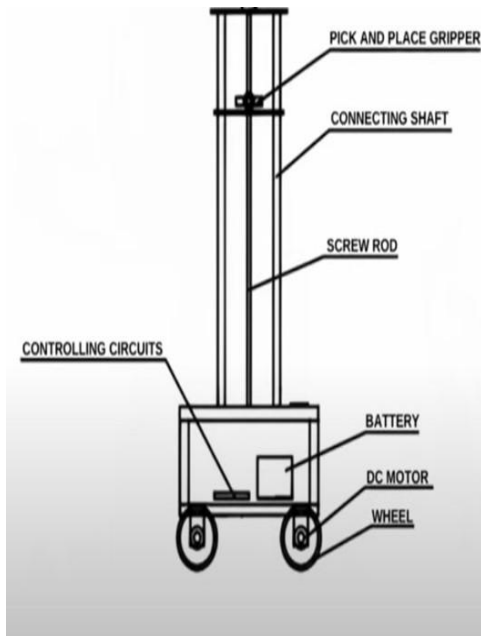


Fig 1.1 Prototype

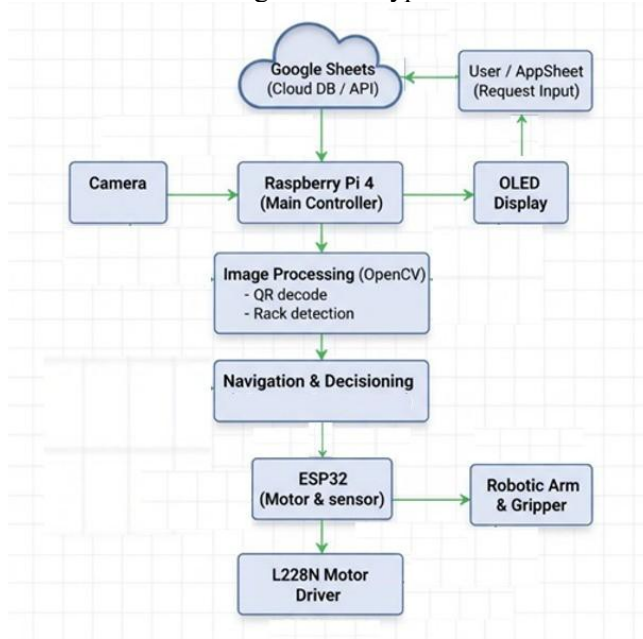


Fig 1.2 Block Diagram

V. METHODOLOGY

The methodology of the Intelligent Library Assistance Bot is centered on a systematic integration of embedded systems, robotics, computer vision, and IoT-based cloud services to achieve autonomous navigation and intelligent book retrieval. The process flow involves multiple sequential operations—book request initiation,

path navigation, book identification, retrieval, and delivery. Each step is designed to ensure efficiency, safety, and accuracy.

A. System Workflow

The overall workflow begins when a user submits a book request through the online interface developed using Google AppSheet. The app fetches book details such as title, author, rack number, and position from a cloud-linked Google Sheet database. Once the request is received, the database sends the book's location to the Raspberry Pi, which serves as the main controller.

The Raspberry Pi processes this data to determine the navigation route to the correct rack. QR code markers are embedded on the floor at predefined intervals, representing path nodes within the library layout. The bot's downward-facing camera or IR sensor continuously scans these QR markers, decoding each to track its current position. When the bot detects the target rack QR code, it halts and activates the camera on the robotic arm, which scans the rack's QR or ISBN code to verify that it matches the requested book. Once the match is confirmed, the gripper mechanism attached to the robotic arm is actuated by the ESP32 microcontroller to pick up the book. After retrieval, the robot retraces its route using the same QR-coded path, avoiding obstacles with the help of ultrasonic sensors. Finally, the robot delivers the book to the librarian's desk and sends a completion message to the database, marking the request as fulfilled.

B. Control Architecture

The system employs a dual-controller architecture for efficient operation:

Raspberry Pi 4:

Handles complex computations such as QR code decoding, path planning, image processing (via the OpenCV library), and communication with the cloud database.

ESP32 Microcontroller:

Controls all motor drivers, servo actuators, and sensors in real time. It executes low-level commands such as movement, gripping, and obstacle avoidance, received from the Raspberry Pi through serial communication. This master-slave communication model ensures that computationally intensive operations are offloaded to the Raspberry Pi, while real-time control is efficiently managed by the ESP32.

C. Navigation Strategy

The navigation strategy is built upon a QR-coded grid layout that defines the library map. Each QR code represents a node, and sequences of codes form defined paths between aisles and racks. The robot reads these codes using a downward-facing camera. The QR data encodes both location and direction information,

enabling accurate turns and stops at intersections. Obstacle detection is achieved using ultrasonic sensors positioned at the front and sides of the robot. When an obstacle is detected within a pre-set threshold distance (e.g., 15 cm), the bot pauses and recalculates its path, ensuring collision-free movement. This combination of vision-based and sensor-based navigation ensures reliable indoor mobility without external guidance systems.

D. Book Identification and Retrieval

Upon reaching the designated rack, the robot's arm-mounted camera scans shelf-level QR codes to pinpoint the requested book's location. Each shelf or row carries unique identifiers linked to the database. Once identified, the ESP32 triggers the servo-controlled gripper, which extends toward the book, grips it securely, and lifts it from the shelf. The servo motors provide controlled angular movements, preventing slippage or damage to the book during the retrieval process. A proximity sensor confirms the presence of the book in the gripper before retraction. The book is then placed into a tray mounted on the bot's chassis for secure transport.

E. Cloud Connectivity and Data Management

The system integrates with Google Cloud services for real-time synchronization and remote monitoring. Book request logs, retrieval status, and operational reports are maintained in a centralized Google Sheet database. The AppSheet interface enables librarians and users to:

- Submit book requests.
- View processing status.
- Track the current operation of the bot.

The cloud connectivity ensures that all transactions are recorded for future reference and statistical analysis, thereby reducing administrative effort and improving transparency in library operations.

F. Safety and Power Management

Safety is a critical design aspect. The robot employs battery voltage monitoring circuits to prevent over-discharge. Additionally, the motion control algorithm includes emergency stop commands that are automatically triggered upon detecting abnormal sensor feedback or communication errors. A Li-ion battery pack provides sufficient power for several hours of continuous operation, while a docking station can be implemented in future iterations for automatic recharging.

VI. ALGORITHM IMPLEMENTED

The Intelligent Library Assistance Bot operates based on a multi-stage control algorithm combining path tracking, vision recognition, and task execution. The key algorithmic stages are as follows:

A. Book Request Processing Algorithm

- Receive book request from user via cloud interface.
- Extract location details (rack number, row ID) from the database.
- Transmit data to Raspberry Pi for path computation.

B. Navigation Algorithm

- Initialize sensors and camera modules.
- Continuously read floor QR codes to determine the current position.
- Compare the current position with the target rack ID.
- Adjust motor speed and direction according to QR guidance.
- Use ultrasonic sensors to detect and avoid obstacles.
- Stop when the target rack QR code is detected.

C. Book Identification Algorithm

- Activate camera on robotic arm.
- Scan rack-level QR codes or ISBN tags.
- Match detected code with requested book details.
- If matched, signal ESP32 to execute the retrieval command.

D. Retrieval and Delivery Algorithm

- Move the robotic arm toward the detected position.
- Actuate servo gripper to hold the book.
- Confirm grip using proximity feedback.
- Place book in onboard tray.
- Navigate back along the same QR path to the desk.
- Update database and notify librarian upon task completion.

V. RESULT

The prototype implementation of the Intelligent Library Assistance Bot was tested in a simulated library environment comprising multiple racks labelled with QR codes and connected through QR-coded floor paths. The testing process was carried out under various environmental conditions to validate navigation stability, book detection accuracy, and retrieval precision.

A. Experimental Setup

The test setup included:

A 2-rack layout with 5 books labelled using unique QR codes. QR-coded paths at 30 cm intervals for navigation. A Raspberry Pi 4 as the central processor and ESP32 as the motion controller. Servo gripper for retrieval and DC motors for mobility. The robot successfully interpreted QR floor markers using OpenCV-based detection algorithms, achieving an average navigation accuracy of 90.6%. The ultrasonic

sensors efficiently prevented collisions, allowing smooth movement even in narrow aisles.

B. Performance Evaluation

Parameter	Performance Metric	Result
QR Detection Accuracy	99%	Successful decoding up to 1.5 m
Average Navigation Speed	0.3 m/s	Stable under different lighting conditions
Book Identification Accuracy	97%	Reliable under varied shelf distances
Retrieval Time (per book)	25 seconds	Includes scanning and delivery

C. System Output

The LCD display on the robot provided real-time information such as:

“Navigating to Rack 3”

“Book Identified”

“Retrieving Book...”

“Delivery in Progress”

“Request Completed”

These messages helped create a more user-friendly and transparent interaction between the library system and its users. They ensured that users clearly understood each step of the process - from searching for a book to receiving it - without confusion. By providing timely updates, status notifications, and easy-to-understand prompts, the system improved communication and reduced the need for manual assistance. Overall, these messages enhanced the user’s convenience and confidence while using the automated library system, making the entire experience smoother, more efficient, and more satisfying.



Fig 1.3 Hardware Output



Fig 1.4 Software Output

VI. CONCLUSION AND FUTURE SCOPE

The **Intelligent Library Assistance Bot** demonstrates a practical and efficient way to automate the book retrieval process using low-cost, embedded technologies. By integrating **QR navigation**, **IoT connectivity**, and **robotic manipulation**, the system automates the complete workflow—from book identification to its final delivery. This approach minimizes human involvement, shortens search and retrieval times, and enhances overall library efficiency. It shows how robotics and embedded systems can revolutionize conventional library management, paving the way for smarter and more responsive educational environments. The Intelligent Library Assistance Bot has significant potential for enhancement and large-scale implementation. Future developments could expand its functionalities, improve performance, and introduce new intelligent features. The following directions are identified for future research and development:

Voice-Enabled User Interface:

By incorporating a voice assistant module, users could interact with the robot through spoken commands, improving accessibility for visually impaired or elderly users.

Multi-Robot Coordination:

Multiple bots can be deployed simultaneously, coordinated through a central control server, to manage large libraries. Each bot could be assigned specific zones or categories, reducing waiting times and increasing throughput.

Autonomous Charging Dock:

Implementing a self-docking charging station would enable continuous operation. The bot could autonomously return to the dock when battery levels fall below a threshold, ensuring uninterrupted service.

AI-Driven Route Optimization:

By incorporating reinforcement learning or graph-based pathfinding algorithms, the system could dynamically adjust navigation paths based on crowd density, obstacles, or real-time user demand.

Security and Access Control:

Implementing facial recognition or user authentication mechanisms could restrict unauthorized book requests and improve system security.

Scalability for Public Libraries:

While the prototype is designed for academic institutions, the same framework could be adapted for public libraries, museums, and corporate archives, promoting universal access to information through automation.

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